

Yulong Zhao

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Education

University of Arizona <i>PhD in Statistics</i>	Tucson, AZ, USA Sept 2023 – May 2028 (Expected)
University of Wisconsin–Madison <i>MS Statistics: Statistics and Data Science (MSDS) Program</i>	Madison, WI, USA Sept 2022 – June 2023
University of Science and Technology of China <i>BS in Mathematics and Applied Mathematics – Statistics</i>	Hefei, Anhui, China Sept 2018 – July 2022

Research Experience

Statistical Methods for Response Process Data University of Arizona <i>Research Assistant, supervised by Prof. Xueying Tang (NSF #2310664)</i>	Sept 2023 – Present
- Developed a scalar-on-process modeling framework for educational process data to identify which actions have time-varying effects and how those effects change over the course of problem solving. - Separated action-frequency effects from smooth timing effects in variable-length sequences using B-spline time bases, adaptive Group LASSO estimation, and bootstrap uncertainty summaries. - Applied the method to the PISA 2012 Tickets item with 19,346 action events from 2,685 students, producing interpretable action-time effect profiles that align with the task's compare-and-commit structure. - Distinguished frequency-only and timing-sensitive actions and used repeated holdout validation as a predictive sanity check, improving mean test accuracy from 93.22% to 96.20% over a frequency-only baseline.	
Emission-Q Recovery for Process-Based Diagnostic Models University of Arizona <i>Q-matrix / hidden Markov diagnostic classification research project</i>	2026 – Present
- Developed a clean framework for estimating emission-Q matrices in process-based diagnostic models from observed action sequences, transition structure, and weak expert anchors. - Built a Python simulation and estimation pipeline for hidden Markov diagnostic classification models, producing Q-matrix estimates, row uncertainty summaries, latent-class posterior diagnostics, and reproducible search records. - Conducted simulation and real-data analyses to study identifiability and failure modes in emission-Q recovery, including signal dilution, state confounding, softmax competition, and nuisance-parameter compensation. - Formalized runtime legality boundaries that separate estimator inputs from oracle information, recovery metrics, and post-hoc diagnostic evidence.	
Mathematical Modeling of Mass Extinction Events University of Arizona <i>Interdisciplinary Research Cluster, Institute for Mathematical and Statistical Innovation (NSF #1929348)</i>	May 2024 – Present
- Built a data pipeline aligning geochemical and environmental proxies ($\delta^{18}\text{O}$, $\delta^{13}\text{C}$, $^{87}\text{Sr}/^{86}\text{Sr}$, temperature, CO_2, sea level) to a unified timescale with missingness and scale harmonization. - Engineered lag, rolling, and change-point features and trained regularized and tree-based models with time-aware validation to support interpretable early-warning analysis.	
COVID-19 Policy Impact Analysis University of Wisconsin–Madison <i>Research Assistant, supervised by Prof. Vivak Patel</i>	Sept 2022 – Jan 2023
- Built linear and hierarchical models linking gathering restrictions to infection rates. - Ran sensitivity analyses to quantify uncertainty sources in policy-effect estimates.	
Probabilistic Forecasting with DeepAR University of Science and Technology of China <i>Research Assistant, supervised by Prof. Yu Chen</i>	June 2021 – Aug 2021
- Implemented an autoregressive recurrent network for probabilistic time-series forecasting. - Developed methodology for multi-series probabilistic prediction.	

Factor-Adjusted Regularized Model Selection | University of Science and Technology of China Sept 2020 – Jan 2021

Research Assistant, supervised by Prof. Canhong Wen

- Identified limitations of standard LASSO under strong cross-sectional and serial dependence.
- Proposed factor-adjusted regularization to address dependence structures.

Conference Presentation

Refereed oral presentation, “Scalar-on-Process Model for Educational Assessment: Modeling Time-Varying Action Effects with Group LASSO Regularization,” Symposium on Data Science & Statistics, Milwaukee, WI. April 30, 2026

Awards and Honors

Research Fellowship, University of Arizona	2024, 2025, 2026
Incoming PhD Student Award, University of Arizona	2023
Academic Excellence Award for MS Statistics, University of Wisconsin–Madison	2022
Outstanding Undergraduate Student Award, University of Science and Technology of China	2018, 2019